

Task 2: Speech-to-Text and Text-to-Speech Module

1. Module Overview

This module is responsible for the voice interaction part of the Smart Cafeteria Self-Checkout Kiosk System.

It contains two ROS Python nodes:

```
nodes/stt_node.py
nodes/tts_node.py
```

The functions are:

```
stt_node.py:
Microphone / keyboard input → text → publish user speech

tts_node.py:
Receive system response text → speak or print response
```

2. ROS Topics

Node	Topic	Type	Direction	Description
stt_node.py	/smart_cafeteria/ user_speech	std_msgs/ String	Publish	Publishes recognized user speech
stt_node.py	/smart_cafeteria/ stt_status	std_msgs/ String	Publish	Publishes STT status
tts_node.py	/smart_cafeteria/ kiosk_response	std_msgs/ String	Subscribe	Receives system response from LLM / Brain node
tts_node.py	/smart_cafeteria/ tts_status	std_msgs/ String	Publish	Publishes TTS status

3. Required Dependencies

For Ubuntu / ROS Noetic:

```
sudo apt update
sudo apt install python3-pip python3-pyaudio -y
pip3 install SpeechRecognition pyttsx3
```

If the microphone is not available, `stt_node.py` will automatically switch to keyboard fallback mode.

4. How to Run

Terminal 1: Start ROS Core

```
source /opt/ros/noetic/setup.bash
roscore
```

Keep this terminal running.

Terminal 2: Run STT Node

```
source /opt/ros/noetic/setup.bash
cd ~/SmartCafeteria
python3 nodes/stt_node.py
```

If microphone is available, speak into the microphone.

Example:

```
done
```

If microphone is not available, the node will show:

```
user_speech>
```

Then type:

```
done
```

and press Enter.

Terminal 3: Check STT Output

```
source /opt/ros/noetic/setup.bash
rostopic echo /smart_cafeteria/user_speech
```

Expected output:

```
data: "done"
```

This means the STT node is publishing user speech correctly.

Terminal 4: Run TTS Node

```
source /opt/ros/noetic/setup.bash
cd ~/SmartCafeteria
python3 nodes/tts_node.py
```

Expected log:

```
TTS Node started.
Subscribed to: /smart_cafeteria/kiosk_response
TTS Node running. Waiting for kiosk responses...
```

Terminal 5: Test TTS Input Topic

```
source /opt/ros/noetic/setup.bash
rostopic pub /smart_cafeteria/kiosk_response std_msgs/String "data: 'Hello, welcome to smart
cafeteria.'"
```

Expected output in the TTS terminal:

```
Received kiosk response: Hello, welcome to smart cafeteria.
```

If audio output is available, the computer will speak the response.

If audio output is not available, the message will still be received and printed in the terminal.

5. Testing Result

The STT topic was tested successfully:

```
/smart_cafeteria/user_speech  
data: "done"
```

The TTS topic was tested successfully:

```
/smart_cafeteria/kiosk_response  
Received kiosk response: Hello, welcome to smart cafeteria.
```

6. Notes

In WSL Ubuntu, real microphone and speaker access may not always work because audio device passthrough depends on Windows and WSL settings.

However, the ROS topic communication has been tested successfully.

For final demonstration, it is recommended to test real microphone and speaker output on a native Ubuntu ROS machine or the lab robot computer.